

Trevor Black

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EDUCATION

Johns Hopkins University

August 2022 – Expected May 2026

Bachelor of Science in Computer Science, GPA 3.6

Baltimore, MD

- Relevant Courses: Machine Learning, Functional Programming, Cryptography, Algorithms, Fullstack JavaScript, Data Structures, Computer Architecture, Networks, Game Development.

EXPERIENCE

Software Development Intern

Summer 2025

Interactive Brokers

Greenwich, CT

- Improved Spring back-end event signalling system; streamlined and modernized old code and design patterns.
- Reduced system's memory usage by over 90% with my new custom Redis cache system I implemented.
- Cache now updates instantly; removed need for slow 5-minute cache refreshes that caused stale data usage.
- Optimized Kafka message consumer from 2 million to 5 million messages processed per minute.

Computer Architecture Course Assistant

2025 – Present

Johns Hopkins University

Baltimore, MD

- Provided official help for hundreds of students in core course on computer OS and architecture fundamentals.
- Graded large coding projects and proctored exams of students.

Coding Instructor

Summer 2024

iD Tech

New York, NY

- Designed and delivered interactive Lua/Roblox coding lessons to teens. Promoted to teach Unreal Engine.

Delineo – Undergraduate Research

Spring 2023

Johns Hopkins University

Baltimore, MD

- Headed Unity sub-team responsible for improving “gamified” version of the pandemic simulation.
- Implemented pandemic event system, handling of server simulation data, and pre-existing bug fixes.

EXTRACURRICULARS AND PROJECTS

Hopkins Design Build Fly Club | President for 2025 competition year

2023 – Present

- Led 3 meetings per week to design and manufacture airplane, handle fundraising, recruitment, etc.
- Increased funding by \$8,000 and more than doubled active member count to 15.
- Placed 49th out of 159 teams (top 1/3) in 2025 competition; improvement from 67th out of 149 (top 1/2) in 2024.
- Led website development, primarily using the Astro web framework, visible here: <https://jhu-dbf.github.io/>

Hopkins Sustainable Wind Energy Team

2022 – 2024

- Competed in Collegiate Wind Competition to build a mini wind turbine to operate under unique conditions.
- Implemented ESP32 PID controller for blade pitch angle to regulate RPM during high-speed winds.

Projects | These projects (and more) detailed on portfolio page: <https://trevoroblack.com/projects>

- **Tail Smash:** Coded a 2D arcade driving game using C++, SFML, Dear ImGui and GLSL. Implemented an Entity Component System to make use of cache locality for boosted performance. Researched and programmed 2D physics to simulate a car and inverse kinematics.
- **ML Language Detector:** Designed and programmed machine learning algorithm to detect language of text using ID3 + Adaboost decision trees. Utilized scikit-learn to learn common bigrams in languages to compare to bigrams common in given text. Achieved 87% accuracy - 12% improvement over single letter frequency analysis.
- **HopHacks Hackathons:** Participated in annual Johns Hopkins hackathons in 2022 and 2023 in team of 4. Created Flutter app incorporating google geofence API to provide local emergency services contacts for users.

SKILLS

Languages: C++, Java, C#, Python, JavaScript, HTML, OCaml, Go, Lua.

Libraries/Technologies: Spring Boot, Kafka, Redis, React, Next.js, Node.js, Unity, pandas, SDL, Git, Unix.

Other (interests): running, skiing, rock climbing, tennis, 3D CAD (Blender, Solidworks), Spanish (conversational).